

Table 1: Table 4: Union Mediation of the Outsourcing Wage Penalty, 2001-2025

Period	Union Rate In-House	Union Rate Out- sourced	Gap	Union Premium (β)	Raw Penalty	Union- Gap Compo- nent (log points)	% Explained	N
Janitors								
2001-2005	20.8%	10.7%	10.1pp	0.167 (0.010)	-0.061 (0.009)	-0.017 (0.002)	27.6%	14,46
2006-2010	20.5%	9.6%	10.9pp	0.151 (0.010)	-0.054 (0.009)	-0.016 (0.002)	30.5%	13,49
2011-2015	18.4%	8.9%	9.5pp	0.159 (0.012)	-0.054 (0.009)	-0.015 (0.002)	28.1%	13,42
2016-2019 [†]	16.9%	8.4%	8.5pp	0.117 (0.012)	-0.026 (0.008)	-0.010 (0.001)	38.0%	9,99
2021-2025	16.4%	9.7%	6.7pp	0.089 (0.013)	-0.033 (0.011)	-0.006 (0.001)	18.2%	8,98
<i>Change, 2001-2005 to 2021-2025</i>					+0.028 (0.014)	+0.011 (0.002)		
Guards								
2001-2005	17.5%	8.3%	9.3pp	0.144 (0.017)	-0.108 (0.010)	-0.013 (0.002)	12.4%	5,42
2006-2010	15.3%	9.1%	6.2pp	0.155 (0.019)	-0.113 (0.013)	-0.010 (0.002)	8.4%	5,53
2011-2015	17.7%	8.5%	9.3pp	0.160 (0.020)	-0.151 (0.013)	-0.015 (0.003)	9.8%	5,41
2016-2019 [†]	19.4%	10.9%	8.5pp	0.144 (0.021)	-0.104 (0.015)	-0.012 (0.003)	11.7%	4,07
2021-2025	17.8%	8.6%	9.2pp	0.050 (0.026)	-0.061 (0.015)	-0.005 (0.003)	7.6%	3,69
<i>Change, 2001-2005 to 2021-2025</i>					+0.047 (0.018)	+0.009 (0.003)		

Gap = union rate, in-house minus outsourced, in percentage points. Union premium (β) is the union coefficient from the author's base model. Raw penalty is the outsourced coefficient from the nonunion model (no union control). Union-gap component (log points) = $-(\text{gap} \times \text{union premium})$, with an approximate delta-method SE: $\sqrt{\beta^2 \text{var}(\text{gap}) + \text{gap}^2 \text{var}(\beta)}$; shown as negative, in the same direction as the raw penalty. % explained = $|\text{union-gap component}| / |\text{raw penalty}|$, and is unstable where the raw penalty is small. The change row compares 2001-2005 to 2021-2025 (independent bins, $\text{SE} = \sqrt{\text{SE}_1^2 + \text{SE}_2^2}$). SEs in parentheses. Descriptive decomposition, not causal. Data: CPS-ORG via IPUMS.